SHUBHAM GAJJAR

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RESEARCH INTERESTS

Vision-language models for biomedical imaging, multimodal contrastive learning, parameter-efficient fine-tuning of large foundation models, attention mechanisms for medical image segmentation and classification, and scalable data engineering for terabyte-scale microscopy and cell-imaging datasets.

EDUCATION

Northeastern University, Portland, Maine — September 2025 – May 2027
Master of Science in Artificial Intelligence, Grade Point Average: 4.0/4.0
Relevant Coursework: Foundations of Artificial Intelligence, Machine Learning, Research Capstone, Algorithms, Applied Programming and Data Processing for Artificial Intelligence

LDRP Institute of Technology and Research, Gandhinagar, India — September 2022 – May 2025
Bachelor of Engineering in Computer Engineering, Grade Point Average: 8.41/10.0
Relevant Coursework: Deep Learning, Computer Vision, Image Processing, Data Structures and Algorithms

VPMP Polytechnic, Gandhinagar, India — September 2019 – May 2022
Diploma in Computer Engineering, Grade Point Average: 9.22/10.0

EXPERIENCE

Northeastern University, Portland, Maine
Research Assistant — February 2026 – Present

Building a vision-language model for automated cell tumor classification from microscopy imaging.

- Designed end-to-end vision-language model fine-tuning pipeline using Google MedGemma 4B (SigLIP vision encoder with Gemma-3 language backbone) and QLoRA, loading the model in 4-bit NF4 quantization with LoRA adapters (r=16, alpha=32, dropout=0.05) on attention projections plus a fully trainable language-model head and embedding layer while freezing the vision encoder
- Engineered preprocessing pipeline merging multi-channel microscopy into normalized RGB composites with per-image P1/P99 normalization resized to 512×512, on a 70:20:10 train, validation, and test split
- Generated a visual question answering dataset across 5 prompt types from expert-reviewed annotations, applying cross-entropy loss on response tokens only through label masking so that roughly 20 percent of tokens contributed to the objective
- Trained on an NVIDIA H100 80GB GPU with effective batch size 8, learning rate 2e-4, gradient clipping, and 2 epochs across approximately 6,000 steps; accelerated data loading through in-memory image caching with batched PySpark binary reads (500 per batch) and ThreadPoolExecutor merging, resolving driver-memory overflow and persisting the cache across subsequent versions
- Evaluated checkpoints every 25 steps on token accuracy, cross-entropy loss, ROUGE-L, and BERTScore; logged metrics, parameters, and training-curve artifacts through MLflow with versioned model artifacts that grew incrementally across iterations

BigCircle (UPSAAS Technologies LLP), Gandhinagar, India
Artificial Intelligence Engineer — January 2025 – August 2025

Built AI-powered research tools and engineered backend systems for content automation at an early-stage product company.

- Developed end-to-end Deep Research pipeline in Python orchestrating large language model-based prompt generation, Firecrawl API for web scraping, OpenAI ChatGPT API for content summarization, and graph visualization with Matplotlib
- Automated Typst PDF report generation and reduced processing time by 75 percent through API call batching and concurrency tuning
- Engineered pagination and authentication systems using JavaScript and Next.js, accelerating page load times by 40 percent

RESEARCH

MorphoCLIP: Cell Microscopy and Text Contrastive Learning — April 2026

Role: Team Lead (3-person team)

Led a 3-person team building a contrastive learning system aligning cell microscopy with natural-language descriptions of biological perturbations (drug treatments, CRISPR knockouts, and Open Reading Frame overexpression) across 3 million-plus images, 303 compounds, and 160 genes from the CPJUMP1 benchmark (Nature Methods, 2024). Achieved 24.3 percent text-to-image Recall at 10 on an 817-perturbation retrieval task, 3.3 times the performance of CellCLIP and 200 times the random baseline, using approximately 8 million trainable parameters versus 1.48 billion (a 185 times reduction); median rank 28 out of 817 (top 3.4 percent). Architected a dual-encoder design pairing a DINOv3 vision backbone (300 million parameters) with a BioClinical ModernBERT text encoder (150 million parameters), a custom CrossChannelFormer fusing 5 fluorescence channels, and Cross-Well Alignment for batch-effect correction integrated into the contrastive objective. Enabled joint training on drug, CRISPR, and Open Reading Frame perturbations, three capabilities absent from prior work. Compressed 3.31 tebibytes of raw microscopy into 153 gigabytes of cached DINOv3 embeddings, reducing per-epoch training time to 3 to 5 minutes on a single RTX 5080 and enabling dozens of ablation configurations per day; managed a 134-commit codebase across 14 feature branches with documentation in Nextra.

A Hybrid ResNet-ViT Architecture for Skin Cancer Classification Published in IEEE, July 2025

Authors: Shubham Gajjar, Om Rathod, Deep Joshi, Harshal Joshi, Vishal Barot

Hybrid architecture combining a frozen ResNet50 backbone as feature extractor with Vision Transformer blocks using four-head multi-head self-attention. Integrated Global Average Pooling and transformer-based global dependency modeling for seven-class skin lesion classification. Achieved 96.3 percent testing accuracy and a macro F1 score of 0.961 on the HAM10000 dataset, with Area Under Curve of approximately 1.00 across all classes (melanoma, nevus, basal cell carcinoma, benign keratosis, dermatofibroma, actinic keratosis, vascular lesions). Applied stratified data augmentation (rotation, flips, brightness/contrast adjustments) to scale 10,015 images to roughly 74,353. Trained with the Nadam optimizer (initial learning rate 0.001) and Sparse Categorical Crossentropy loss on a stratified 70/15/15 split. Published at IEEE 4th World Conference on Applied Intelligence and Computing, July 2025. DOI: 10.1109/AIC66080.2025.11212073.

Extended ResNet50: Inverse Soft Mask Attention for Skin Cancer Classification May 2025

Authors: Shubham Gajjar, Om Rathod, Deep Joshi, Harshal Joshi, Vishal Barot

Two-stage deep learning pipeline integrating U-Net++ hair segmentation with an Extended ResNet50 classifier featuring a novel Inverse Soft Mask Attention mechanism. Implemented dense residual blocks and Squeeze-and-Excitation modules with learnable weighted feature aggregation combining hair-occluded and unoccluded image regions. Achieved 97.89 percent testing accuracy and 97.74 percent validation accuracy at epoch 22 on HAM10000 (10,015 dermoscopic images across seven classes). Trained with Nadam, Cosine Decay Restarts, and Sparse Categorical Crossentropy. Across 21 architectural trials evaluating Vision Transformers, hybrid models, and custom attention mechanisms, the model outperformed SCCNet (95.20 percent), VCCINet (93.18 percent), and SPCB-Net (97.10 percent).

VGG16-MCA UNet: A Hybrid Deep Learning Approach for Enhanced Brain Tumor Segmentation in FLAIR MRI August 2024

Authors: Shubham Gajjar, Deep Joshi, Avi Poptani, Vishal Barot

Hybrid segmentation framework integrating a pretrained VGG16 encoder with a Multi-Channel Attention decoder for brain tumor segmentation in FLAIR MRI scans. Applied Focal Tversky Loss to address severe class imbalance where tumor regions cover only 2 to 5 percent of image area. Implemented ensemble learning combining predictions from multiple checkpoints and architectural configurations, with skip connections, batch normalization, and dropout for regularization. Achieved 99.59 percent accuracy and 99.71 percent specificity on the LGG Brain MRI Segmentation dataset (110 low-grade glioma patients from the TCGA collection). Trained with Adam (initial learning rate 0.05), ReduceLROnPlateau, and EarlyStopping. Conducted 35 systematic experiments against UNet, Attention UNet, and Scaler Attention UNet baselines. Preprocessing included skull stripping, intensity normalization, and resizing to 256 by 256 pixels, achieving superior Dice coefficient and Intersection over Union scores.

CONFERENCE PRESENTATIONS

Gajjar, S., Rathod, O., Joshi, D., Joshi, H., & Barot, V. (2025, July). *A Hybrid ResNet-ViT Architecture for Skin Cancer Classification.* Oral presentation, IEEE 4th World Conference on Applied Intelligence and Computing, presented to an audience of more than 100 attendees.

PROJECTS

Next-Step: Full-Stack School Management Platform May 2026

Full-stack school management platform with async REST API and cross-platform mobile app for administrators, teachers, and students.

- Built async REST API with FastAPI exposing 20-plus endpoints across 10 router modules backed by 9 SQLAlchemy ORM models with relationship mapping, cascade deletes, JSON columns, and unique constraints; managed via Alembic migrations on PostgreSQL with auto-generated Swagger and ReDoc documentation
- Implemented JWT authentication using Argon2 password hashing and Google OAuth 2.0 with PKCE flow, supporting Admin, Teacher, and Student roles
- Crafted cross-platform mobile application (iOS, Android, Web) using React Native and Expo with TypeScript, Zustand state management, file-based routing via Expo Router, and AsyncStorage session persistence
- Integrated Google Drive API for document management and set up Celery with Redis (Flower monitoring) for background task processing of notifications and asynchronous operations
- Co-developed within a 5-person team using feature-branch Git workflow with 11-plus pull requests, code review, pytest with httpx for asynchronous integration testing, and AI-assisted development tools for prototyping and review

Reinforcement Learning Agent for TrackMania November 2024

- Built an autonomous racing agent in PyTorch implementing Implicit Quantile Networks with NumPy-based state processing, reaching 85 percent track completion through iterative reward shaping
- Optimized the training loop with parallel environment simulations on graphics processing units accelerated by CUDA, reducing training time by 60 percent

TECHNICAL SKILLS

Programming Languages: Python, JavaScript
Deep Learning: PyTorch, TensorFlow, Keras, HuggingFace Transformers, CUDA (GPU acceleration), QLoRA
Computer Vision and Multimodal: OpenCV, Albumentations, Matplotlib, MedGemma, MedSigLIP, DINOv3, CLIP
Data Science and MLOps: NumPy, Pandas, Scikit-learn, MLflow, PySpark, Docker, PostgreSQL
Web Development and Tools: Next.js, React Native, FastAPI, Tailwind CSS, Git

CERTIFICATIONS

Python for Data Science, Indian Institute of Technology Madras, 2024
Python Data Structures, University of Michigan (Coursera), 2024